

Prototype a Grounded Assistant

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of capstone grounded assistant.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

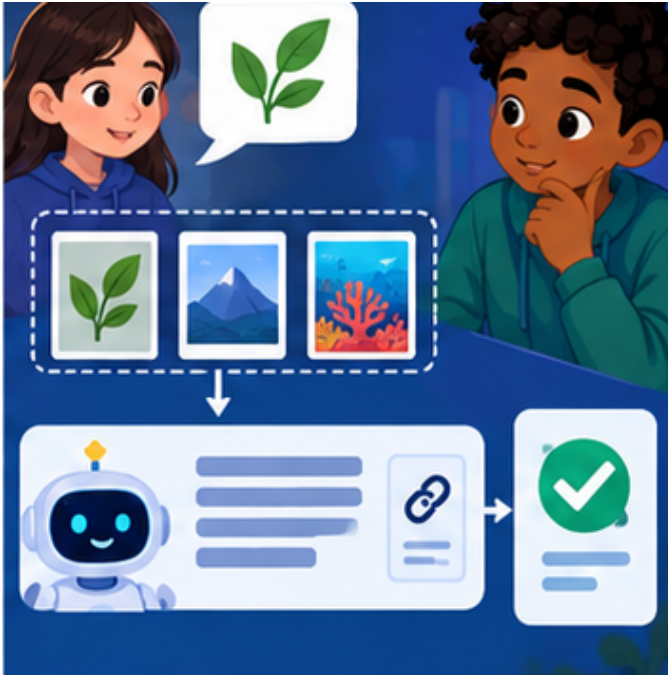
Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 24: Prototype a Grounded Assistant

A smallest working product connects prompt, references, interaction, and output.



CREATOR MISSION

Build one end-to-end Study Buddy path.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Capstone requirements and corpus. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

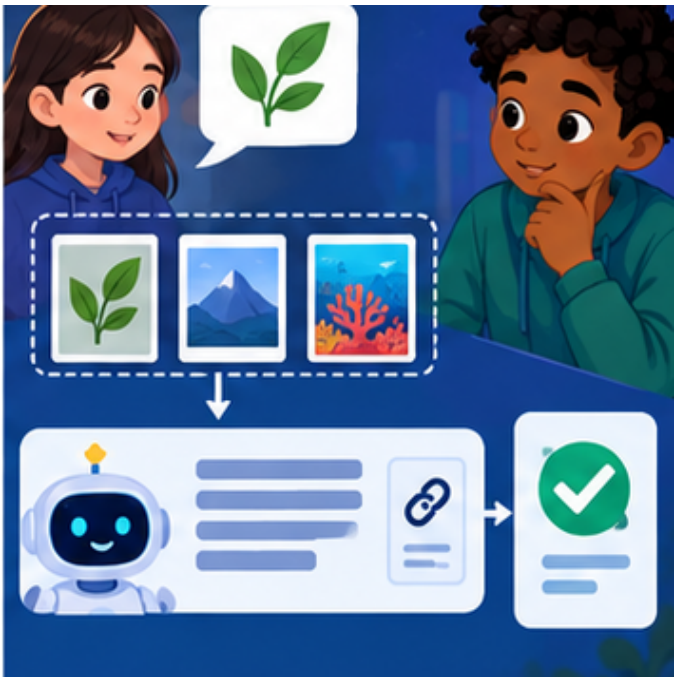
This week's end goal

By the end, I can explain prototype a grounded assistant using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use prototype a grounded assistant to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Build the assistant as a sequence of testable stages: validate request, retrieve authorised evidence, decide answerability, draft, validate and return. Avoid merging all errors into one model score.

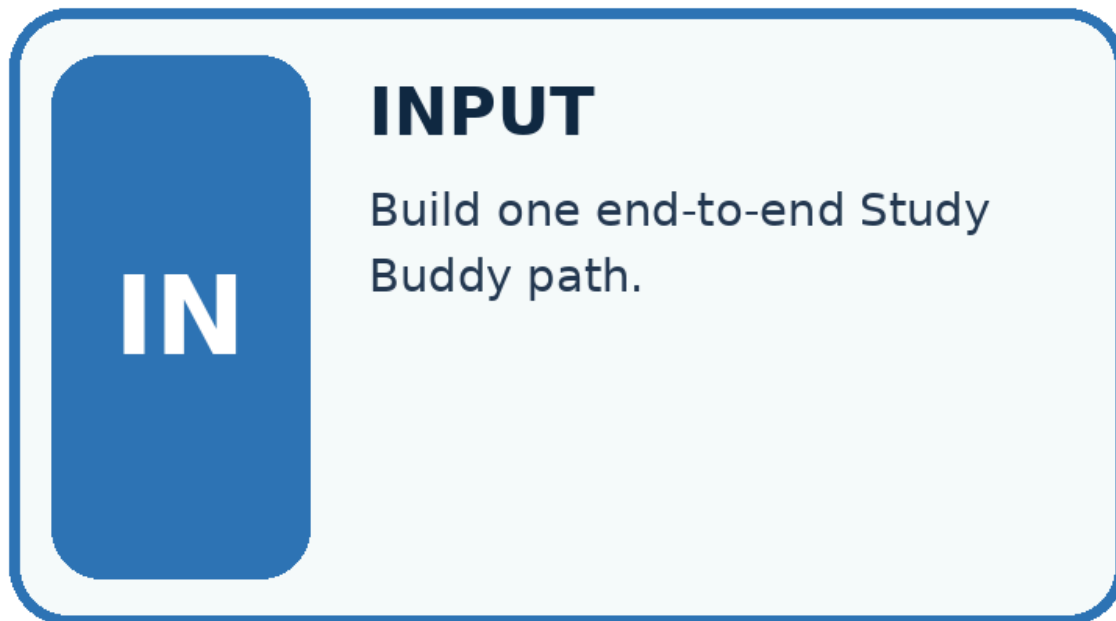
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

A smallest working product connects prompt, references, interaction, and output.



OUT

EVIDENCE

Build one end-to-end Study Buddy path.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.

Capstone grounded assistant — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.

Check or calculation: A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.

Conclusion and limit: Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: PL3 is absent while PL2 remains.

Prediction: Explain that the daily check is not specified in the available source.

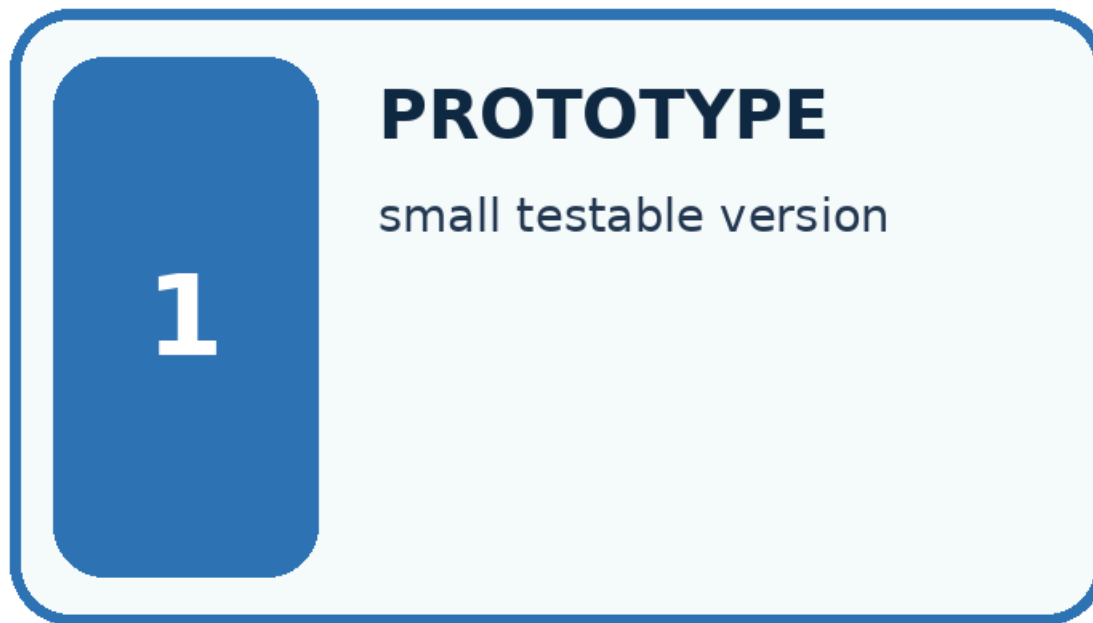
Reason: The remaining general water fact does not establish this activity's schedule.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

END-TO-END

complete path from input to output

3

GROUNDING

answer tied to approved references

4

FALLBACK

safe behavior when a stage fails

Word · My own example · Not this / boundary

Word: prototype

My own example: An early working or simulated version used to test assumptions about a design.

Not this / boundary: _____

Word: end-to-end

My own example: Covering the entire process from original input to final output.

Not this / boundary: _____

Word: grounding

My own example: Connecting an answer or action to specified evidence or observations.

Not this / boundary: _____

Word: fallback

My own example: A predefined alternative action used when the usual path cannot be completed reliably.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Pipeline stage

Meaning: One operation with defined inputs and failure handling

Word: Answerability gate

Meaning: A check for sufficient evidence before answering

Word: Semantic validator

Meaning: A check that claims match task evidence

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 24 • WORKED EVIDENCE

- 1 Input: learner question plus selected course notes.
- 2 Retrieve and choose a relevant passage.
- 3 Generate structured answer and citation; validate fields and support.
- 4 Missing evidence triggers ask-more or 'not found in sources.'

1. Input: learner question plus selected course notes.
 2. Retrieve and choose a relevant passage.
 3. Generate structured answer and citation; validate fields and support.
 4. Missing evidence triggers ask-more or 'not found in sources.'
- Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.

Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prototype a Grounded Assistant

Build a minimal grounded study answer with checks between retrieval and display.

The facts we will use

Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Prototype a Grounded Assistant

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieve evidence

PL3 directly supports the daily soil-check claim; PL2 supplies a different fact.



2. Match question

Choose the activity's soil check, not a general watering recommendation.



3. Validate draft

Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

CHANGE ONE THING

PL3 is absent while PL2 remains.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Retrieve evidence

What happens and why: PL3 directly supports the daily soil-check claim; PL2 supplies a different fact.

Step: 2. Match question

What happens and why: Choose the activity's soil check, not a general watering recommendation.

Step: 3. Validate draft

What happens and why: Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A grounded prototype still needs an evaluation set and permission checks before anyone relies on it.

Read the labelled picture: Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: PL3 is absent while PL2 remains. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A prototype is complete when the happy path works once. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project,

select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A correct happy path does not establish fallback, access control or side-effect safety.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone grounded assistant”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN
Build one full path, then test normal, ambiguous, missing-source, and malformed-output cases; record stage failures.

Build one full path, then test normal, ambiguous, missing-source, and malformed-output cases; record stage failures.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Connect user interface, source store, retrieval, prompt, schema validation, and evidence display.

THIS WEEK’S CHECKPOINT Connect user interface, source store, retrieval, prompt, schema validation, and evidence display.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Build one end-to-end Study Buddy path.

CHALLENGE Build one end-to-end Study Buddy path.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Approved notes say plants need light and roots take up water. Ask how roots help.

The question

Specify the minimum grounded path.

Worked explanation

Retrieve the roots note, answer that roots take up water and reference it. Validate support and avoid unrelated claims.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Build the smallest grounded path

Evidence or stage · Value or result

Evidence or stage: Question

Value or result: How do roots help?

Evidence or stage: Retrieve

Value or result: Roots take up water

Evidence or stage: Answer

Value or result: Explain water uptake

Evidence or stage: Check

Value or result: Reference and verify the roots note

Approved notes say plants need light and roots take up water. Ask how roots help.

Specify the minimum grounded path.

Retrieve the roots note, answer that roots take up water and reference it. Validate support and avoid unrelated claims.

14. Try a new case independently

Independent case: Apply a new case independently

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources.

Trace stages and distinguish zero from missing.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 24 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A study assistant has a supplied passage about roots. A user asks a question about planets. What should the prototype do within a passage-only boundary?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Connect user interface, source store, retrieval, prompt, schema validation, and evidence display.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or fallback.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Build a minimal grounded study answer with checks between retrieval and display.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of capstone grounded assistant

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of capstone grounded assistant

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or

assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.